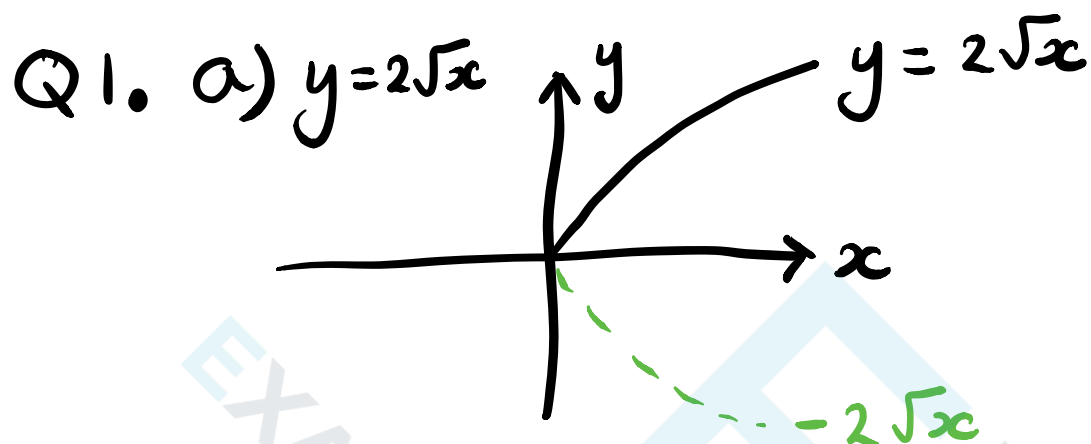
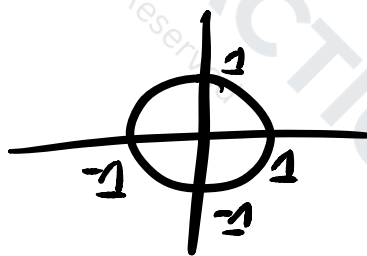


Conic graphs

Basic skills



Q2. a) $x^2 + y^2 = 1$



b) $\frac{x^2}{4} + \frac{y^2}{9} = 1$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

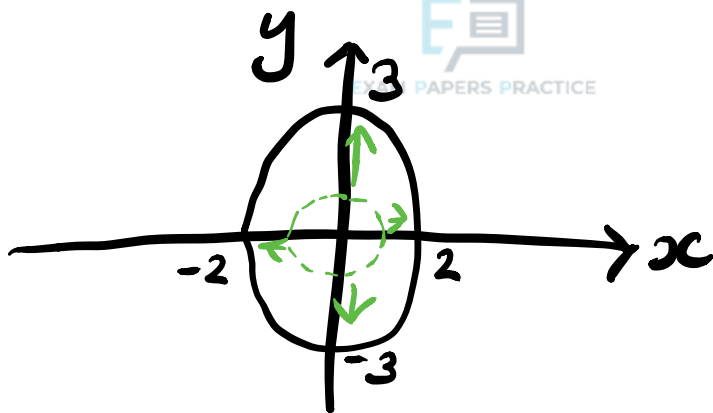
$$x \mapsto \frac{x}{2}$$

SF: 2 in x -dir

$$y \mapsto \frac{y}{3}$$

SF: 3 in y -dir

c)



Q3. a) $\frac{x^2}{2^2} - \frac{y^2}{3^2} = 1$ $a=2, b=3$

so $y = \pm \frac{3}{2}x$
 $\pm \frac{b}{a}x$

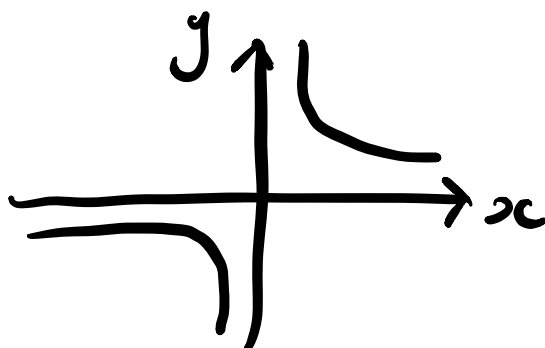
b)



At $y=0$ $x^2=2^2$ $x=\pm 2$

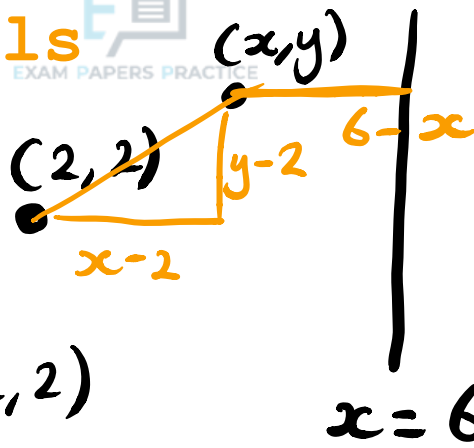
Q4. $xy=4$

i.e.
 $y = \frac{4}{x}$



Competency skills

Q5.



a) (x, y) and $(2, 2)$
distance

$$\sqrt{(x-2)^2 + (y-2)^2}$$

b) (x, y) and $x=6$ line is
 $\leftarrow$ if $x < 6$ $6-x$ or $x-6$ $\leftarrow$ if $x > 6$

i.e. $|6-x|$ or $\sqrt{(6-x)^2}$

useful

c) so

$$\sqrt{(x-2)^2 + (y-2)^2} = 2\sqrt{(6-x)^2}$$

$$\Rightarrow (x-2)^2 + (y-2)^2 = 4(6-x)^2$$

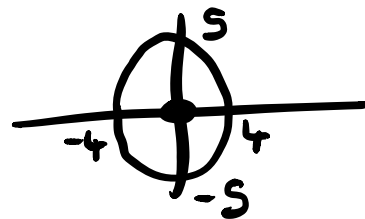
$$\Rightarrow x^2 - 4x + y^2 - 4y + 8 = 4x^2 - 48x + 144$$

$$\Rightarrow 3x^2 - 44x - y^2 + 4y + 136 = 0$$

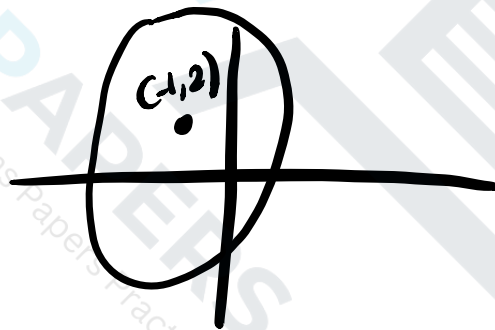
↑ hyperbolic as " $x^2 - y^2$ "

Q6. a) $\frac{(x+1)^2}{16} + \frac{(y-2)^2}{25} = 1$

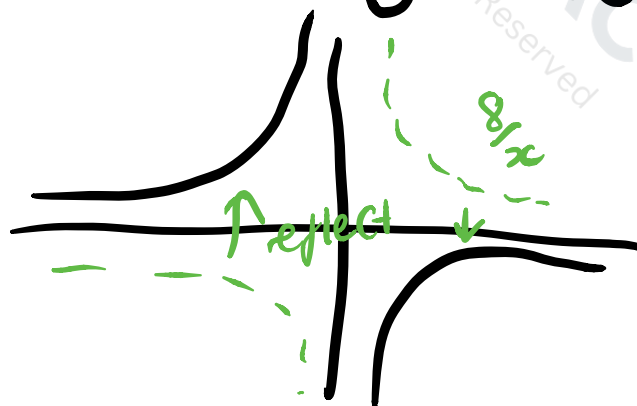
from $\frac{x^2}{4^2} + \frac{y^2}{5^2} = 1$



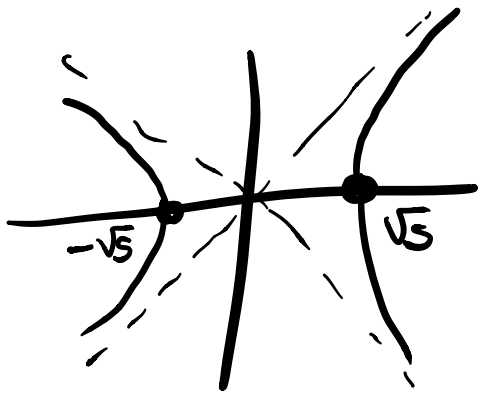
Translate $x \mapsto x+1$ $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$
 $y \mapsto y-2$



b) $xy = -8$ $y = -\frac{8}{x}$



c) $\frac{x^2}{5} - \frac{(2y-5)^2}{25} = 1$ $\frac{\frac{x^2}{5}}{(\sqrt{5})^2} - \frac{y^2}{5^2} = 1$



asymptotes

$$y = \pm \frac{5}{\sqrt{5}} x$$

$$= \pm \sqrt{5} x$$

Then $y \mapsto 2y - 5$

So $y \mapsto y - 5 \mapsto 2y - 5$

$$\begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$SF: \frac{1}{2}$$



Q7. $\frac{x^2}{16} - y^2 = 1 \quad x \mapsto \frac{x}{5}$

$$\frac{\left(\frac{x}{5}\right)^2}{16} - y^2 = 1 \Rightarrow \frac{x^2}{400} - y^2 = 1$$

Then $y \mapsto -y$ $\frac{x^2}{400} - (-y)^2 = 1$

$$\frac{x^2}{400} - y^2 = 1$$

Q8. $y^2 = 4ax$ to pass $(-3, 0)$
need translation $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

$$y^2 = 4a(x+3)$$



Advanced skills

Q9. $\frac{x^2}{5} - \frac{y^2}{2} = 2$

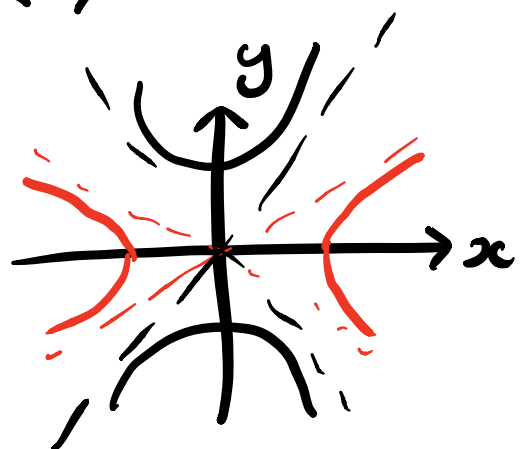
Asymptote
 $y = \pm \frac{2}{5}x$

In line $y = x$

$y \leftrightarrow x$

$$\frac{y^2}{5} - \frac{x^2}{2} = 2$$

Asymptotes
 $y = \pm \frac{5}{2}x$



Q10. $y^2 = 2x$

$\begin{bmatrix} -2 \\ 3 \end{bmatrix}$ $x \mapsto x+2$ $y \mapsto y-3$ $(y-3)^2 = 2(x+2)$

x SF: $\frac{1}{2}$ $x \mapsto \frac{x}{\frac{1}{2}} = 2x$ $(y-3)^2 = 2(2x+2)$

$$(y-3)^2 = 4x + 4$$

Q11.

$$xy = 8, \quad \frac{x^2}{4} - y^2 = 1$$

$$y = \frac{8}{x}$$

$$\frac{x^2}{4} - \frac{64}{x^2} = 1$$

$$\frac{x^4}{4} - 64 = x^2$$

$$x^4 - 4x^2 - 256 = 0$$

$$z^2 - 4z - 256 = 0 \quad z = x^2$$

$$z = \frac{4 \pm \sqrt{1040}}{2}$$

$$= 2 \pm 2\sqrt{65}$$

$$x = \pm \sqrt{2 \pm 2\sqrt{65}}$$

Q12. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\left(\frac{x}{a}\right)^2 - \left(\frac{y}{b}\right)^2 = 1$$

$$\left(\frac{x}{a} + \frac{y}{b}\right) \left(\frac{x}{a} - \frac{y}{b}\right) = 1$$

At large x (and therefore large y)
the 1 is insignificant so

$$\left(\frac{x}{a} + \frac{y}{b}\right) \left(\frac{x}{a} - \frac{y}{b}\right) = 0$$

$$\text{so } \frac{x}{a} \pm \frac{y}{b} = 0 \Rightarrow y = \pm \frac{b}{a} x$$

Q13. $y^2 + (3x-2)^2 = 25$

is $\frac{x^2}{5^2} + \frac{y^2}{5^2} = 1$ then $x \mapsto 3x-2$
 $[-2]$, SF: $\frac{1}{3}$ in x

